

Assignment 04: Employee Attrition Analysis

Dataset Information

Dataset: IBM HR Analytics Employee Attrition & Performance

Kaggle Link: <https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset>

About the Dataset

This is a fictional dataset created by IBM data scientists to understand factors that lead to employee attrition.

Key Information:

- 1,470 employees
- 35 features including demographics, job details, satisfaction scores
- **Target:** Attrition (Yes/No) - whether employee left the company

Accessing the Dataset

You can either download the dataset manually from Google Drive to use in your local Jupyter Notebook, or use the following code to download and load it automatically:

```
import kagglehub
import pandas as pd

# Download the dataset
path = kagglehub.dataset_download("pavansubhasht/ibm-hr-analytics-attrition-dataset/versions/1")
print("Dataset downloaded to:", path)

# Load the dataset into a DataFrame
df = pd.read_csv(f"{path}/WA_Fn-UseC_-HR-Employee-Attrition.csv")
```

Your Task

Create **ONE** Jupyter notebook that performs the following analysis:

1 Data Loading & Basic Information

- Load the dataset.
- Display basic info: `df.info()`, `df.describe()`, `df.head()`, `df.tail()`.
- Check data types.
- Check target variable distribution (Attrition).

2 Data Quality Checks

- Check for missing values.
- Check for duplicate rows.
- Check for any data inconsistencies.

3 Data Cleaning

- Remove duplicates (if any).
- Handle missing values (if any).
- Remove unnecessary columns (e.g., `EmployeeNumber`, `EmployeeCount`).
- Fix any data issues you find.

4 Exploratory Data Analysis (EDA)

Create visualizations to answer:

- What is the age distribution of employees?
- What is the monthly income distribution?
- Which departments have the most employees?
- Which department has the highest attrition rate?
- What is the attrition rate by gender?
- Do employees who left have different income than those who stayed?
- How does age relate to attrition?
- What are the correlations between numerical features?
- Any other interesting patterns you discover.

Requirement: Minimum 5 visualizations required.

5 Data Transformation

a) Train/Test Split

- Separate features (X) and target (y).
- Split into 80% train, 20% test.
- Use stratified split to maintain class distribution.

b) Encoding Categorical Variables

Apply appropriate encoding:

- **Label Encoding** for binary categories (e.g., Gender).
- **One-Hot Encoding** for nominal categories (e.g., Department, JobRole).

c) Feature Scaling

- Identify numerical features that need scaling.
- Apply `StandardScaler`.
- Fit on training data, transform both train and test.

d) Handle Class Imbalance

- Calculate the imbalance ratio.
- Calculate class weights manually using the formula:

$$\text{weight} = \frac{n_samples}{n_classes \times n_samples_per_class}$$

- Verify with sklearn's `compute_class_weight`.

6 Final Summary

Write a comprehensive summary (markdown cell) that includes:

Key Findings:

- What are the 5 most important insights about employee attrition?
- Which factors seem most related to attrition?
- Which employees are most at risk?

Data Preparation Summary:

- Original dataset shape.
- Final dataset shape after cleaning and encoding.
- Class imbalance ratio.
- Class weights calculated.

Recommendations:

- Based on your analysis, what 3-5 recommendations would you make to reduce attrition?

Note

The tasks listed below are requirements, not a strict sequence. You may perform them in whatever order makes logical sense for your analysis — what matters is that your final notebook covers all items.

Submission Requirements

- One Jupyter notebook (.ipynb file).
- Well-commented code - explain what each section does.
- Clear visualizations - with titles and labels.
- Written insights - don't just show numbers, explain what they mean.

Good luck!